

HOW DNA TESTING IS IMPACTING LIVES: A CONSUMER CASE STUDY

Consumer problem	The solution	Results
Sunita Reddy, a teacher from Hyderabad, was suffering from obesity for almost 15 years. She had undergone treatments under various health experts, including cosmetic surgeries to get rid of the excess fat, but all efforts failed. She was constantly looking for a viable solution to her health problem.	Following the advice of a friend in the US, she tried testing her DNA with one of the kits from a leading lab in India to know about the possible causes of obesity stored in her DNA. She sent her saliva in the manner prescribed in the kit for investigation to the lab.	She was amazed with the results. The results showed how obesity was a result of her gene sequencing. Post-result counselling helped her to understand the changes required to be made in her lifestyle (dietary and exercise) to lose the excess body weight.

Business segmentation

Those willing to enter the market can either start as an independent Indian brand provider or serve as Indian partner of a foreign brand.

The Indian providers set up their own brand and independent company. Such companies may also invest in research. These either manufacture their own kits or, as in most cases, import in bulk from China and sell in India under their own brand name.

Indian partners of a foreign brand take exclusive partnership rights to sell DNA testing kits of established brands from the west. They focus on sales. Research and manufacturing is taken care of by the parent company. Such businesses offer high returns on low investment.

Whether you choose to create your own brand or partner with a foreign brand, you'll require channel partners across business services to help in research and interpretation of results.

How to get started?

A minimum of three million rupees (can go much higher depending on the business plan) is required to set up a DNA testing kits business. The investment amount includes the costs for infrastructure establishment, IT and automation. It takes 3-5 years to break even depending upon the competition in market and

your business identity as an Indian brand importing DNA kits from China or the subsidiary of a western counterpart. Currently, direct-to-consumer kits start at around ₹ 2000 and cost as much as ₹ 200,000 for specific tests.

A major capital investment is required to set a corporate facility and getting trained manpower on board. Another major requirement is a profitable marketing and distribution channel powered by the right logistics. Companies like Quick source DNA use LED lighting solutions from Philips to save power costs. XCode uses artificial intelligence solutions offered by Amazon.

DNA testing kit providers need to conform to various ISO quality standards. While organisations seeking to assure responsible environment management system should get ISO 14000 certified, some also oblige to ISO 9001 to assure high-quality practices. Companies that provide a complete solution with their own DNA testing laboratories need to be ISO 17025 certified. For companies providing DNA kits commercially, the formal guidelines aren't yet clear. For other research, some healthcare categories and forensics, specific guidelines prescribed by the concerned agencies need to be followed.

The DNA testing kits market is huge and highly untapped. Thus

there is scope for enormous growth. Cost to start the business is relatively low and return margins are huge. A DNA kit costing ₹ 100 to a dealer might be sold in the market at a price of ₹ 1000," says S. Balaji of Geneagic Testing Kits.

Diagnostic labs are aggressively looking for suppliers of human DNA testing kits since the demand for direct-to-consumer business is growing exponentially, says a spokesperson from Dr Lal Path-Labs. They have tied up with Map My Genome to source their DNA testing kits and further sell these to their channel partners.

Western parent companies (especially from USA and Europe) expect their Indian partners to set up a basic facility running on energy saving resources like solar panels and LEDs because they believe in responsible business management.

Profitability is ambiguous since the business territory is new. Also, the path to reach the untapped end consumers isn't yet defined. The sole source being relied upon is the Internet. There is intense pressure on the firms to conform to regulatory requirements. Also, the standardisation process is highly dubious.

DNA testing business is all about utilising biotechnology to analyse people's DNA sequence for interpretations related to ancestry, identifying health related risk factors and learning about individual physical body characteristics. Currently, the results are being harnessed by health, sports, beauty and lifestyle industry segments. The Internet of Things (IoT) and artificial intelligence are contributing significantly in the collection and analysis process.

Pro Gen Bio Lab: A Business Case

Pro Gen Bio Lab is a B2B company providing DNA testing kits of